

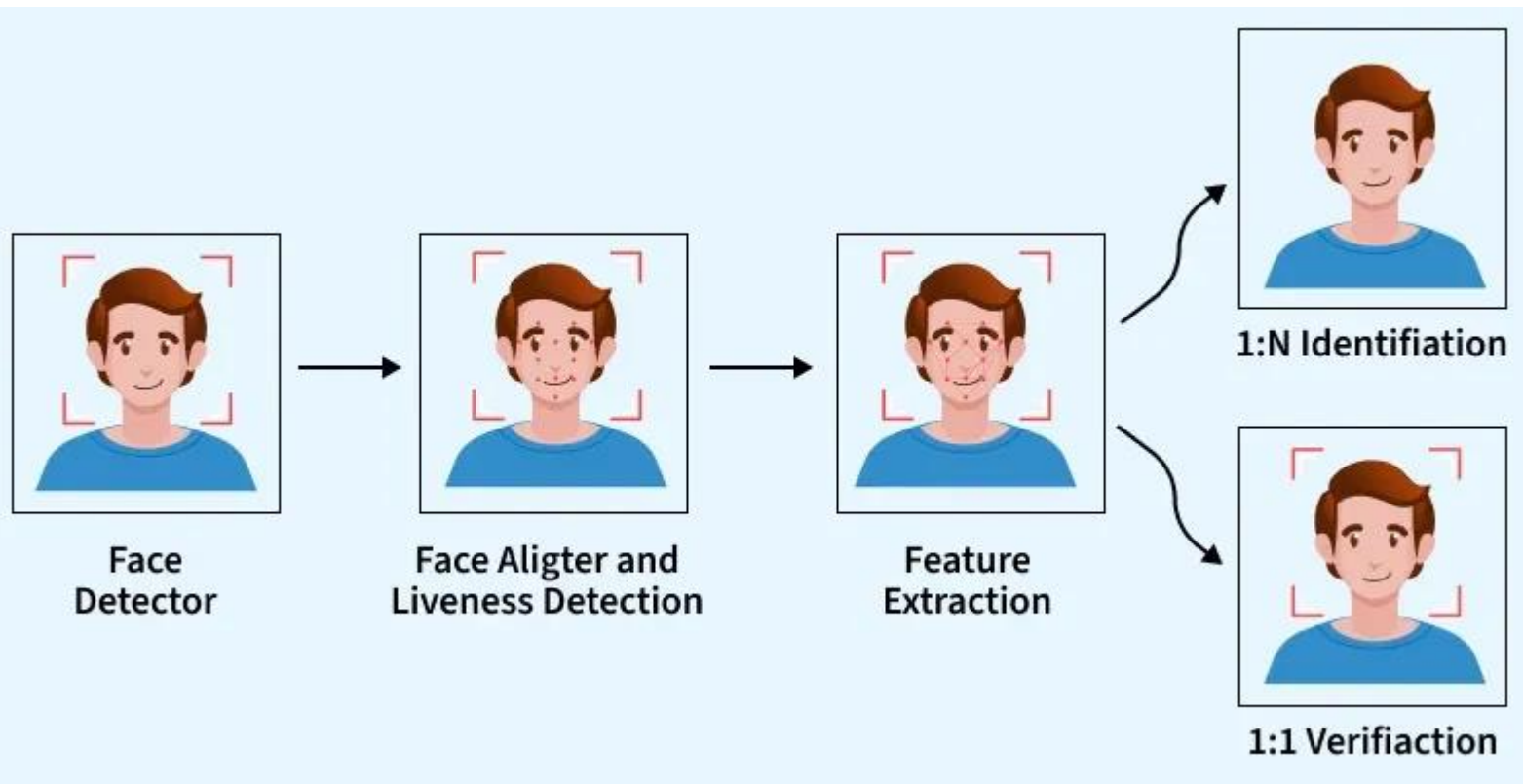


Feature Detection in Digital Image Processing (DIP)

- Feature detection in digital image processing is the process of identifying important and distinctive parts of an image that can be used for analysis, recognition, or further processing. These features represent meaningful information in an image such as edges, corners, blobs, and keypoints.

Why Feature Detection is Important

- **Feature detection helps computers:**
- Understand the structure of an image
- Recognize objects and shapes
- Match images (e.g., in face recognition or image stitching)
- Track objects in videos
- Perform image registration and 3D reconstruction



Face Detection

- ▶ **Face Detection**
- ▶ The first step identifies the region where a face is present in an image or video frame. Popular algorithms include:
- ▶ First, the system finds a face in an image.
- ▶ It separates the face from the background.
- ▶ If there are multiple people, it detects all faces.
- ▶ Common methods:
- ▶ Haar Cascade
- ▶ HOG (Histogram of Oriented Gradients)
- ▶ Deep learning–based detectors (CNNs)

Face Alignment

Once a face is detected, the system aligns it by adjusting key facial landmarks such as the eyes, nose, and lips. Alignment helps handle variations caused by rotation, tilt, lighting or facial expressions, ensuring that the model works on a normalized and correctly oriented face.

After detecting the face, the system:

- Aligns it properly (straightens the face)

- Locates key facial landmarks such as eyes, nose, and mouth

- This step improves accuracy by handling:

 - Head tilt

 - Different face angles

Feature Extraction

- ▶ The system extracts unique facial features.
- ▶ These are mathematical values, not images
- ▶ Features include distance between eyes, nose shape, jawline, etc.
- ▶ Modern systems use deep learning to convert a face into a numerical vector called a face embedding.
- ▶ Popular models:
 - ▶ FaceNet
 - ▶ DeepFaceArcFace

Face Matching / Comparison

- ▶ The extracted features are compared with faces stored in a database.
- ▶ The system calculates similarity
- ▶ If similarity is above a threshold → match found
- ▶ Two cases: Verification (1:1) – Is this person who they claim to be?
- ▶ Identification (1:N) – Who is this person among many?

Decision Making

- ▶ Based on similarity score:
- ▶ Match → access allowed / identity confirmed
- ▶ No match → access denied / unknown person

Common Types of Features

- ▶ **Edges**
- ▶ Edges are locations in an image where there is a sudden change in intensity or color.
- ▶ Represent object boundaries
- ▶ Detected using gradient-based methods
- ▶ **Common edge detectors:**
- ▶ Sobel
- ▶ Prewitt
- ▶ Roberts
- ▶ Canny (very popular and accurate)

Corners

- ▶ Corners are points where two or more edges meet.
- ▶ Highly distinctive and stable features
- ▶ Useful for object recognition and image matching
- ▶ Common corner detectors:
 - ▶ Harris Corner Detector
 - ▶ Shi–Tomasi Detector

Blobs

- ▶ Blobs are regions in an image that differ in brightness or color compared to surrounding areas.
- ▶ Useful for detecting objects or regions
- ▶ Blob detection methods:
 - ▶ Laplacian of Gaussian (LoG)
 - ▶ Difference of Gaussian (DoG)
 - ▶ Determinant of Hessian

- ▶ Keypoints (Interest Points)
- ▶ Keypoints are specific points in an image that are invariant to changes such as scale, rotation, and illumination.
- ▶ Used in advanced computer vision tasks Popular keypoint detectors and descriptors:
- ▶ SIFT (Scale-Invariant Feature Transform)
- ▶ SURF (Speeded-Up Robust Features)
- ▶ ORB (Oriented FAST and Rotated BRIEF)

Applications of Feature Detection

- ▶ Face and object recognition
- ▶ Medical image analysis
- ▶ Remote sensing and satellite imagery
- ▶ Robotics and autonomous vehicles
- ▶ Image stitching and panorama creation